

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

BTC CORPORATION,
Petitioner,

v.

TG BIOTECH CO., LTD.,
Patent Owner.

IPR2022-00998
Patent 8,357,786 B2

Before ULRIKE W. JENKS, JAMES A. WORTH, and
RYAN H. FLAX, *Administrative Patent Judges*.

JENKS, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Granting Patent Owner's Motion to Strike
37 C.F.R. 42.64

I. INTRODUCTION

A. *Background and Summary*

In this *inter partes* review, BTC Corporation (“Petitioner”) challenges claims 1–13 (14–17 are not challenged) of U.S. Patent No. 8,357,786 B2 (Ex. 1001, “the ’786 patent”) assigned to TG Biotech Co., Ltd. (“Patent Owner”). Paper 2 (“Pet.”).

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision, issued pursuant to 35 U.S.C. § 318(a), addresses issues and arguments raised during the trial in this *inter partes* review. For the reasons discussed below, we determine that Petitioner has failed to prove by a preponderance of the evidence that claims 1–13 of the ’786 patent are unpatentable (*see* 35 U.S.C. § 316(e) (2018) (“In an *inter partes* review instituted under this chapter, the petitioner shall have the burden of proving a proposition of unpatentability by a preponderance of the evidence.”)).

B. *Procedural History*

On May 5, 2022, Petitioner filed its Petition seeking *inter partes* review of claims 1–13 of the ’786 patent. On August 17, 2022, Patent Owner filed a Preliminary Response (*see* Paper 6, “Prelim. Resp.”). Pursuant to 35 U.S.C. § 314, we instituted *inter partes* review of challenged claims 1–13 of the ’786 patent on November 15, 2022 (Paper 7, “Decision” or “Dec.”).

After institution of trial, Patent Owner filed a Response (Paper 20, “PO Resp.”), to which Petitioner filed a Reply (Paper 25, “Reply”), and Patent Owner, in turn, filed a Sur-Reply (Paper 32, “Sur-Reply”).

With the Board’s authorization, Patent Owner also filed a Motion to Strike Exhibit 1038. Paper 35 (citing Ex. 3003, “Motion”).

A prehearing conference call was held on August 18, 2023 to discuss various matters. Paper 38. We note that Petitioner has withdrawn its unpatentability Ground E, based on Huh. *Id.* (citing PO Resp. 60 (citing Ex. 3002, 3 (“Petitioner withdraws Ground E of the Petition and will not pursue this ground any further.”); *see* Pet. 64–67.

Oral argument was held on August 31, 2023. A transcript of the oral argument is included in the record. (Paper 43, “Hearing Trans.”).

C. *Real Parties in Interest*

Petitioner identifies itself, BTC Corporation, as the real party-in-interest. Pet. 4. Patent Owner identifies itself, TG Biotech Co., Ltd., as the real party-in-interest. Paper 4, 2.

D. *Related Matters*

Petitioner states that “[t]here are no judicial or administrative matters in the United States that would affect or be affected by a decision in this proceeding.” Pet. 4. Patent Owner agrees. Paper 4, 2.

E. *Asserted Grounds of Unpatentability*

Petitioner contends that claims 1–13 of the ’786 patent are unpatentable on the following grounds:

Claim(s) Challenged ¹	35 U.S.C. § ²	Reference(s)/Basis
1–13	103(a)	Kim 460, ³ Razmovski ⁴
1–13 ⁵	103(a)	Kim 460, Kang, ⁶ Baek ⁷
8–10, 12, 13	103(a)	WO 178, ⁸ Razmovski
1–13	103(a)	WO 178, Kang

Pet. 6. Petitioner relies on, *inter alia*, the Declarations of Neal Davies, Ph.D. (Ex. 1003; Ex. 1035) and Seung Su Lee, Ph.D. (Ex. 1013; Ex. 1017). Patent

¹ In the Petition, Petitioner asserted one additional ground directed to claims 8, 9, and 12 based on US Patent Pub. No. 2006/0246163 A1, published November 2, 2006 (Ex. 1020) (“Huh”). Pet. 64–67. That ground has been withdrawn by Petitioner. *See* Ex. 3002, 3.

² The Leahy-Smith America Invents Act (“AIA”) included revisions to 35 U.S.C. § 103 that became effective on March 16, 2013. Because the ’786 patent issued from an application filed before March 16, 2013, we apply the pre-AIA versions of the statutory bases for unpatentability.

³ US Patent No. 5,776,460, issued July 7, 1998 (Ex. 1005). The parties refer to this reference as “Kim 460;” we do the same.

⁴ Razmovski-Naumovski, V., et al., *Chemistry and pharmacology of Gynostemma pentaphyllum*, PHYTOCHEMISTRY REVIEWS 4:197–219 (2005) (Ex. 1007). The parties refer to this reference as “Razmovski;” we do the same.

⁵ In the “Grounds of Challenge” table, it appears that Petitioner inadvertently switched Ground B with Ground C. *See* Pet. 6, 41, 47.

⁶ Korean Pub. No. 10-2006-0063041, published June 12, 2006. We refer to the English-language translation of Kang provided by Petitioner (Ex. 1010).

⁷ Korean Pub. No. 10-2007-0109282, published November 15, 2007. We refer to the English-language translation of Baek provided by Petitioner (Ex. 1008).

⁸ International Pub. No. WO 2008/140178 A1, published November 20, 2008 (Ex. 1006). The parties refer to this reference as “WO 178;” we do the same.

Owner relies upon, *inter alia*, the Declaration of Valentina Naumovski, Ph.D. Ex. 2011.

F. *The '786 patent (Ex. 1001)*

The '786 patent is titled “Method for Preparing *Gynostemma Pentaphyllum* Extract with Increasing Damulin A and Damulin B Contents, and Pharmaceutical Compositions of the Same for Treating Metabolic Disease.” Ex. 1001, code (54). The '786 patent “relates to a method for preparing a novel composition-containing *Gynostemma pentaphyllum* extract, in which a *Gynostemma pentaphyllum* extract is treated with high temperature and high pressure, thereby increasing contents of damulin A and damulin B (active ingredients for AMPK [(AMP-activated protein kinase)] activation)” and “a method for using a novel *Gynostemma pentaphyllum* extract with a novel composition having increased contents of damulin A and damulin B to efficiently treat and improve metabolic syndrome[s] including obesity, diabetes, hyperlipidaemia, or the like.” *Id.* at 1:19–29.

The '786 patent explains that AMPK is a heterotrimer protein that exists generally in muscle cells in abundance and “plays an important role in appetite regulation, weight regulation, blood glucose regulation, and blood lipid metabolism regulation, or the like.” *Id.* at 1:31–38. The '786 patent further explains that *Gynostemma pentaphyllum* is a perennial vine of the family *Cucurbitaceae* and an extract tea derived from such has been used for a variety of ailments, such as constipation, diarrhea, asthma, and bronchitis, and provided health benefits, including restoring internal organ functions, maintaining healthy skin, and relieving pain and ulcers from stress. *Id.* at 2:27–41. It is known that “a *Gynostemma pentaphyllum* extract can improve metabolic syndrome, including insulin resistance, obesity, and

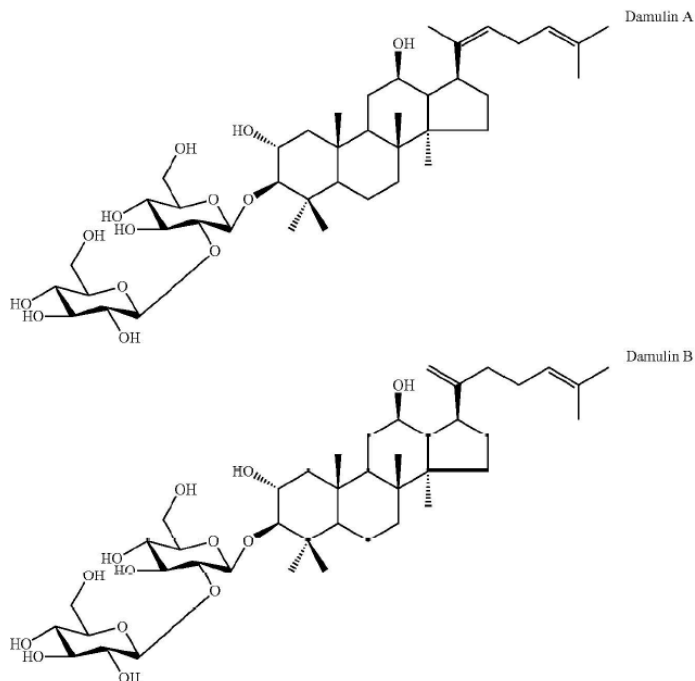
hyperlipidaemia, by increasing the activity of AMP-activated protein kinase (AMPK).” *Id.* at 2:42–45.

The ’786 patent purports to have found, “through the analysis of the components of the *Gynostemma pentaphyllum* extract” “that damulin A and damulin B included in the *Gynostemma pentaphyllum* extract are influential in the activity of AMPK” and provides a “novel *Gynostemma pentaphyllum* extract with a novel composition having increased contents of damulin A and damulin B” that “was determined to have a better capability for phosphorylation and enzymatic activation of AMPK, than a conventional *Gynostemma pentaphyllum* extract.” *Id.* at 2:46–56.

The ’786 patent discloses that

2 α ,3 β ,12 β -trihydroxydammar-20(22)-E,24-diene-3-O-[β -D-glucopyranosyl-(1 $\rightarrow$)- β -D-glucopyranoside] named “damulin A” and 2 α ,3 β ,12 β -trihydroxydammar-20,24-diene-3-O-[β -D-glucopyranosyl-(1 $\rightarrow$)- β -D-glucopyranoside] named “damulin B”, which are novel dammarane-based saponins and have a capability for enzymatic activation of AMP-activated protein kinase (AMPK), are separated and analyzed from a *Gynostemma pentaphyllum* extract by using HPLC and NMR. The damulin A and the damulin B have the following structures.

[Formula 1]



Id. at 3:3–48. The process of separating the damulins from a *Gynostemma pentaphyllum* extract and the preparation of the *Gynostemma pentaphyllum* extract include extracting from dried *Gynostemma pentaphyllum* leaves using an ethanol aqueous solution having a volume 10 to 20 times as large as the volume of the leaves, taking out a first supernatant, adding an ethanol aqueous solution (6 to 20 times), taking out a second supernatant through a 1 to 8 hour extraction at 65–95° C, putting the first and second supernatants together and filtrating to result in an extracted product, concentrating to provide a *Gynostemma pentaphyllum* extract concentrate (referred to as “TG1022”), and then treating the TG1022 with a high temperature and a high pressure to provide a *Gynostemma pentaphyllum* extract concentrate having increased contents of damulins (referred to as “TG1022F”). *Id.* at 3:49–4:3.

In its “EXAMPLE 1,” the ’786 patent discloses separation of damulin as an AMPK activating material from a *Gynostemma pentaphyllum* extract,

in which “5 kg of dried *Gynostemma pentaphyllum* leaves was dipped in 50 l[iters] of 50% ethanol, and then, after [a] first extraction at 90° C. for 6 hours, a first supernatant was collected” and to “the residual ethanol-dipped material of the *Gynostemma pentaphyllum* leaves, 50 l[iters] of 50% ethanol was again added, and then a second supernatant was collected through 6 hour extraction at 90° C.” *Id.* at 7:35–46. In its “EXAMPLE 6,” the ’786 patent discloses preparation of *Gynostemma pentaphyllum* extract having increased damulin contents through treatment of high temperature and high pressure, in which the “*Gynostemma pentaphyllum* extract (TG1022F) was prepared by performing a high temperature/high pressure reaction or a ultra-high pressure reaction on the conventional *Gynostemma pentaphyllum* extract (TG1022) obtained by ethanol extraction at room temperature and atmospheric pressure” and then in a high temperature/high pressure reaction, “a conventional high temperature/high pressure sterilizer (autoclave) or a ultra-high pressure reactor” is used. *Id.* at 12:15–45. Table 2 of the ’786 is reproduced below.

TABLE 2				
Reaction conditions			Content of damulins %	
temperature	Time	Pressure	(w/w)	
(° C.)	(Hour)	(atmosphere)	damulin A	damulin B
Control group (TG1022)			0.37 ± 0.03	0.28 ± 0.02
121	0.5	1.2	0.70 ± 0.04	0.56 ± 0.04
	1		0.89 ± 0.07	0.68 ± 0.04
	2		1.53 ± 0.06	1.16 ± 0.07
	3		2.01 ± 0.08	1.47 ± 0.08
	4		2.49 ± 0.09	1.81 ± 0.10
	8		3.21 ± 0.03	2.42 ± 0.03
	12		4.51 ± 0.03	4.12 ± 0.02
	24		6.82 ± 0.06	5.54 ± 0.05
45	12	297	0.62 ± 0.03	0.52 ± 0.02
		691	0.83 ± 0.07	0.74 ± 0.07
		1087	1.21 ± 0.06	1.13 ± 0.06
	24	297	0.82 ± 0.02	0.71 ± 0.05
		691	1.22 ± 0.04	0.97 ± 0.03
		1087	1.81 ± 0.06	1.33 ± 0.02

Id. at 13:1–20. Table 2 shows various conditions for producing *Gynostemma pentaphyllum* extracts, which indicate an increase of damulin content in the

Gynostemma pentaphyllum extract when high temperatures and high pressures are used. *Id.* at 12:65–67.

G. *The Challenged Claims*

Claims 1–13 are the challenged claims in this proceeding. *See* Section I.D. above. Claims 1, 8, and 10 are the independent claims challenged by Petitioner in this proceeding. Independent claims 1, 8, and 10, reproduced below, are illustrative of the claimed subject matter:

1. A method for preparing a novel *Gynostemma pentaphyllum*⁹ extract having increased concentrations of damulin A and damulin B, comprising the step of:

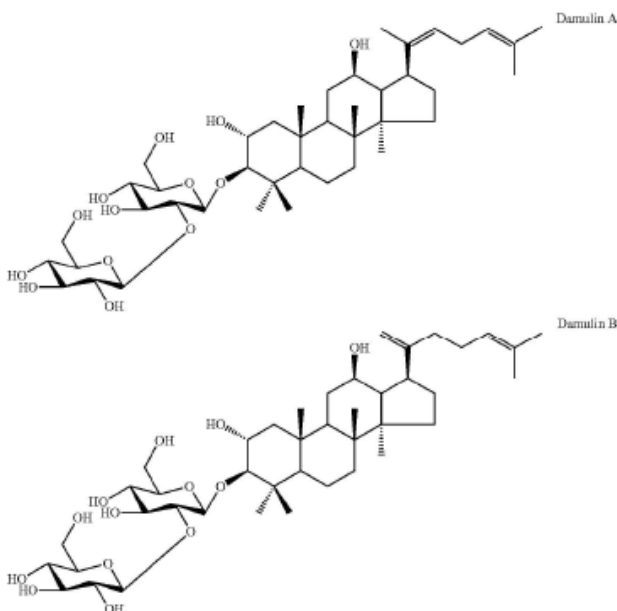
subjecting an ethanol extract of *Gynostemma pentaphyllum* leaves to a temperature of about 40-125° C. and a pressure of about 1.2~1100 atmospheres for 1-24 hours to obtain increased concentrations of damulin A and damulin B, which are 0.7-7% (w/w) and 0.5-6% (w/w), respectively, based on the total weight of a dry powder of the novel *Gynostemma pentaphyllum* extract.

Ex. 1001, 16:60–17:2.

8. A pharmaceutical composition for improving and treating a metabolic syndrome, comprising at least one of at least 0.7% damulin A and at least 0.5% damulin B, represented by Formula 1:

⁹ *Gynostemma pentaphyllum*, *G. pentaphyllum* and GP are used interchangeably throughout this decision.

[Formula 1]



and a pharmacologically acceptable carrier or excipient, wherein the metabolic syndrome is obesity, diabetes, or hyperlipidaemia.

Id. at 17:25–18:3.

10. A novel *Gynostemma pentaphyllum* extract comprising 0.7-7% (w/w) of damulin A and 0.5-6% (w/w) of damulin B based on the total weight of a dry powder of the *Gynostemma pentaphyllum* extract.

Id. at 18:7–10.

II. ANALYSIS

A. *Principles of Law*

“In an IPR, the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim”)). This burden of persuasion never shifts to Patent Owner. *See*

Dynamic Drinkware, LLC v. Nat’l Graphics, Inc., 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review).

To prevail in its challenge to Patent Owner’s claims, Petitioner must demonstrate by a preponderance of the evidence¹⁰ that the claims are unpatentable. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d). A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time of the invention to a person having ordinary skill in the art. *KSR Int’l Co. v. Teleflex, Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including “the scope and content of the prior art”; “differences between the prior art and the claims at issue”; and “the level of ordinary skill in the pertinent art,” and any evidence of non-obviousness, if presented. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In analyzing the obviousness of a combination of prior art elements, it can be important to identify a reason that would have prompted one of skill in the art “to combine . . . known elements in the fashion claimed by the patent at issue.” *KSR*, 550 U.S. at 418. A precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *Id.* Rather, “any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a

¹⁰ The burden of showing something by a preponderance of the evidence requires the trier of fact to believe that the existence of a fact is more probable than its nonexistence before the trier of fact may find in favor of the party who carries the burden. *Concrete Pipe & Prods. of Cal., Inc. v. Constr. Laborers Pension Tr. for S. Cal.*, 508 U.S. 602, 622 (1993).

reason for combining the elements in the manner claimed.” *Id.* at 420. Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (citations omitted).

B. *Level of Ordinary Skill in the Art*

The level of ordinary skill in the art is a “lens” through which we view the prior art and the claimed invention. *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001). The person of ordinary skill in the art is a hypothetical person presumed to have known the relevant art at the time of the invention. *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995). In determining the level of ordinary skill in the art, we may consider certain factors, including the “type of problems encountered in the art; prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active workers in the field.” *Id.*

Petitioner¹¹ asserts that a person of ordinary skill in the art would have had “an undergraduate science degree and would also have a Ph.D. in

¹¹ Petitioner contends that we should give Dr. Naumovski’s testimony little weight because “Dr. Naumovski lacks a basis for this opinion. Dr. Naumovski testified that she had never had responsibilities in a lab that was developing pharmacological agents from biological sources.” Reply 18, (citing Ex. 1036, 21:15–19). “After completing her Ph.D. in 2005, she took a several years’ leave of absence. . . . Since returning to work in 2010, she has

chemistry, pharmaceutical science, food science, or chemical engineering, and several years of experience with extraction and analysis of pharmaceutically active materials from plant-based materials in an industrial or academic setting” and “would [have] be[en] familiar with the structure and utility of saponins, and generally aware of the scientific literature related to the extraction and pharmaceutical utility of gypenosides from GP and ginseng.” Pet. 13 (citing Ex. 1003 ¶ 18).

Patent Owner disagrees with Petitioner’s definition of the person of ordinary skill. PO Resp. 11. According to Patent Owner, “a POSA would have had an advanced degree, for example a Ph.D., in chemistry, biochemistry, pharmaceutical science, or a related field, or a Bachelor’s or

primarily been a lecturer and not working in a lab or doing development.” Reply 21 (citing Ex. 1036, 120:16–20).

Patent Owner responds that Petitioner’s reply “comprises *ad hominem* attacks on Naumovski, . . . despite Naumovski authoring the publication Petitioner relies on for Grounds A and B, and despite Naumovski’s numerous publications about *Gynostemma pentaphyllum* (“GP”) outnumbering GP publications by Davies and Lee, combined.” Sur-Reply 4. “Petitioner fails to address Naumovski’s evidence of 1) impacts of purity level on concentration calculations and 2) availability of pure DB standards.” Sur-Reply 3 (citing Ex. 2015 ¶¶ 165–166, 150—51), *see also id.* at 31 (Dr. “Naumovski has published many times about GP including at least eight journal articles, one book chapter, one invited commentary, and five conference presentations during 2005-2010. Ex. 2015, Appendix A (publications 57–61, 63, 64, 68, Book Chapter publication 5, conference presentations 85-89)— more GP-related publications than Davies and Lee combined.”).

We are not persuaded by Petitioner’s contention that we should disregard Dr. Naumovski’s testimony because she may have taken a leave of absence from work for a couple of years. The record supports that Dr. Naumovski not only meets the requirements of the person of ordinary skill in the art, but also appears to have relevant knowledge with respect to *Gynostemma pentaphyllum* (“GP”) saponins. *See* Ex. 2015 (Appendix A).

Master's degree in chemistry, biochemistry, pharmaceutical science, or a related field, and at least two years of experience using techniques for extraction and characterization of plant-based materials." PO Resp. 11 (citing Ex. 2015 ¶ 74). Because there are more than 400 species of plants that contain saponins, Patent Owner disagrees that a person of ordinary skill in the art would be familiar with the structure and utility of saponins. *Id.*

The level of skill in the art is a reference point for gauging what would have been obvious based on the reference disclosures. *See Okajima*, 261 F.3d at 1355. In other words, the lens of the level of ordinary skill in the art does not change a prior art's disclosure, though it may change one's understanding of what is disclosed. Patent Owner, however, does not explain how its definition provides a reference point that would meaningfully alter the Petition's assertions of obviousness.

For purposes of this decision, we analyze the asserted prior art with respect to the level of skill set forth by Petitioner, but there would be no difference in the outcome of this Decision if we were to apply the level of skill in the art set forth by Patent Owner.

C. *Claim Construction*

In an *inter partes* review for a petition filed on or after November 13, 2018, "claim[s] of a patent . . . shall be construed using the same claim construction standard that would be used to construe the claim[s] in a civil action under 35 U.S.C. [§] 282(b), including construing the claim[s] in accordance with the ordinary and customary meaning of such claim[s] as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent." 37 C.F.R. § 42.100(b); *see* Changes to the Claim Construction Standard for Interpreting Claims in Trial Proceedings Before

the Patent Trial and Appeal Board, 83 Fed. Reg. 51,340 (Oct. 11, 2018) (codified at 37 C.F.R. § 42.100(b) (2019)) (amending 37 C.F.R. § 42.100(b) effective November 13, 2018). A term’s ordinary and customary meaning “is its meaning to the ordinary artisan after reading the entire patent.”

Phillips v. AWH Corp., 415 F.3d 1303, 1321 (Fed. Cir. 2005) (en banc).

Nevertheless, “it is always necessary to review the specification to determine whether the inventor has used any terms in a manner inconsistent with their ordinary meaning,” because “[t]he specification acts as a dictionary when it expressly defines terms used in the claims or when it defines terms by implication.” *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996).

Petitioner requests construction of the terms “an effective amount for improving or treating the metabolic syndrome,” which is found in claims 11–13, as meaning “an amount that produces any measurable amount of improvement in the metabolic syndrome” (Pet. 13–15 (citing Ex. 1003 ¶ 91)) and “[a] pharmaceutical composition for improving and treating a metabolic syndrome, comprising at least one of at least 0.7% damulin A and at least 0.5% damulin B . . . and a pharmacologically acceptable carrier or excipient,” which is found in claim 8, as meaning “a pharmaceutical composition that includes an extract with 0.7% and 0.5% of damulin A and damulin B[, respectively,] along with a carrier or excipient” (*id.* at 15–16 (citing Ex. 1002, 64–65, 251–52, 280; Ex. 1001, 12:15–67, Table 2)).

Patent Owner argues that “constructions are not necessary.” PO Resp. 11–12.

We agree with Patent Owner and determine that it is not necessary for purposes of this Decision to construe any claim terms. *See Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999) (only terms

that are in controversy need to be construed, and only to the extent necessary to resolve the controversy).

D. *Overview of Asserted References*

1. *Kim 460 (Ex. 1005)*

Kim 460 “relates to a processed ginseng product with enhanced pharmacological effects and to a drink composition containing the processed ginseng product.” Ex. 1005, 1:7–10. Kim 460 explains that ginseng is known to have various pharmacological effects ranging from preventing aging to providing anti-diabetic effects. *Id.* at 1:23–30.

Kim 460 discloses enhancing these pharmacological effects by way of “a heat-treatment of a ginseng at a high temperature of 120° to 180° C. for 0.5 to 20 hours so as to make a ratio of ginsenoside (Rg₃+Rg₅) to (Rc+Rd+Rb₁+Rb₂) of above 1.0.” *Id.* at 1:10–16. Fresh ginseng is used, which can be *Panax ginseng*, and preferably, the treatment is at a temperature of 120° to 140° C. for 2 to 5 hours, but the heating time can vary depending on the heating temperature. *Id.* at 3:13–25. To increase efficiency, the process may be performed in an airtight container such as an autoclave with the ginseng being soaked in water. *Id.* at 3:29–34.

Alternatively, the processed ginseng may be extracted by using a solvent and then removing the solvent by concentrating or freeze-drying to obtain a dried powder form. *Id.* at 3:41–47. The solvent can be water, a lower alcohol such as methanol or ethanol, a lower ketone, or a mixed solvent. *Id.* at 3:48–51. The processed ginseng product can be used as a variety of medicinal agents or can be incorporated as a food product. *Id.* at 4:34–44.

Kim 460's Table 1 is reproduced below.

TABLE 1							
Relative Peak Area of Saponin Components of Ginseng							
Sample	Saponin						$(Rg_3 + Rg_5) / (Rc + Rd + Rb_1 + Rb_2)$
	Rb ₁	Rb ₂	Rc	Rd	Rg ₃	Rg ₅	
130° C/2 hrs	4.06	3.44	3.98	3.82	21.00	16.17	2.43
120° C/3 hrs	10.82	6.91	8.52	7.64	28.01	14.01	1.24
120° C/2 hrs	10.46	7.74	10.66	5.58	24.12	11.35	1.03
110° C/2 hrs	19.02	11.55	10.68	8.37	7.02	4.39	0.23
White Ginseng	22.15	3.25	5.86	2.47	0.00	0.00	0.00
Red Ginseng	30.11	9.69	12.45	1.76	1.05	1.05	0.01

Id. at 6:5–26. Table 1 shows amounts of ginseng saponin components (ginsenoside Rb₁, Rb₂, Rc, Rd, and Rg₃) obtained from various samples of ginseng some of which were heat treated and the ratios of $(Rg_3 + Rg_5) / (Rc + Rd + Rb_1 + Rb_2)$. *Id.* at 5:35–6:26.

Kim 460's Table 2 is reproduced below.

TABLE 2									
Change in Contents of Ginsenoside Rg ₃ and Rg ₅ Due to a Change in heating temperature									
Ginsenoside	Untreated Ginseng	Heating Temp. (°C.) Times (hrs)							
	(Fresh Ginseng)	100 (2)	110 (2)	120 (2)	130 (2)	150 (2)	160 (2)	180 (2)	200 (2)
Rg ₃	0.00	0.02	0.08	0.17	0.86	0.44	0.45	0.35	0.23
Rg ₅	0.00	0.02	0.05	0.08	0.44	0.53	0.56	0.48	0.38

Note) Contents of each components are expressed as % by weight based on the amount of a fresh ginseng.

Id. at 6:50–63. Table 2 shows the amounts of saponin components of ginsenoside Rb₃ and Rb₅ in ginseng samples that were heated at 100°, 110°, 120°, 130°, 150°, 160°, 180°, and 200°.

120°, 130°, 150°, 160°, 180° and 200°C. for 2 hours in comparison to ginseng that was not heat-treated (“fresh ginseng”). *Id.* at 6:37–63.

2. *Razmovski (Ex. 1007)*

Razmovski is titled “Chemistry and pharmacology of *Gynostemma pentaphyllum*.” Ex. 1007, 1. Razmovski discloses that “[t]he dammarane saponins isolated from *Gynostemma pentaphyllum*, namely gypenosides or gynosaponins, are believed to be the active components responsible for its various biological activities and reported clinical effects” and that “*Gynostemma pentaphyllum* may complement the popular herbal medicine, *Panax ginseng*, as it also contains a high ginsenoside content and exhibits similar biological activities.” *Id.* (Abstr.).

According to Razmovski, “*G. pentaphyllum* (usually dried material) can be extracted with a variety of solvents, including water, methanol and ethanol for the more polar material; ethyl acetate, petroleum ether and chloroform for the less polar material, using an open system or under reflux.” *Id.* at 3. Over 100 saponins have been isolated and identified from *G. pentaphyllum*. *Id.* Of those, eight saponins are the same as protopanaxadiol-type ginsenosides Rb₁, Rc, Rb₃, Rd, F₂, Rg₃, malonyl-Rb₁, and malonyl-Rd that are found in *P. ginseng*, and together, these ginsenosides make up around 25% of the total gynosaponin in the plant. *Id.* at 4. “Total saponin content is reported to be about 2.4% of the dried herb.” *Id.* Razmovski provides Table 1 listing “Saponine structures [R1, R2, R3, R4, R5, R6, and R7] of the dammarane type in *G. pentaphyllum*.” *Id.* at 5–9.

Razmovski discloses a study showing that *G. pentaphyllum* has a “lipid lowering effect” in rats and quails. *Id.* at 4. Another study shows that in rats, *G. pentaphyllum* extract reduced serum triglyceride (TG) by 53% and

by 85% in a 4 days treatment and in a 12 days treatment, respectively. *Id.* at 10–11. Clinical studies also show beneficial blood lipid change in hyperlipidaemic patients. *Id.* at 11–12. Other studies show that the saponin from *G. pentaphyllum* improves sensitivity to insulin receptors in diabetic rats. *Id.* at 10–11.

3. *Kang (Ex. 1010)*

Kang is titled “Composition Containing Ginsenoside Rg5 and Rk1 for improvement of Brain Function and Cognitive Function.” Ex. 1010, code (54). Kang explains that “[g]insenoside Rg3 (R), Rg3 (S), Rg5, and Rk1 are present in trace amounts in red ginseng and are known to have anti-epileptic effects.” *Id.* at 3. Kang discloses that ginsenoside Rg5 and Rk1 may be purified and separated from the roots of Panax plants containing ginseng saponin, such as *Panax ginseng*, or *Gynostemma pentaphyllum* which contains “a compound of the dammaran[e] backbone, and the like, leaves of these plants, tissue cultures of these plants, or extracts thereof with water or lower alcohols.” *Id.* at 4.

Kang discloses a ginsenoside Rg5 and Rk1-containing pharmaceutical composition that “comprises 0.1 wt% to 50 wt% of the compounds based on the total weight of the composition” and discloses that the composition may further include “suitable carriers, excipients and diluents typically used in preparation of the pharmaceutical composition.” *Id.* at 4–5.

In Kang’s “Example 1,” a composition containing ginsenoside Rg5 and Rk1 is prepared by obtaining a methanol extract through reflux extraction of ginseng in methanol via heating at 130°C for 2 hours. *Id.* at 7. This example produces 20 g of a fraction containing ginsenoside Rg5 and Rk1. *Id.* In Kang’s “Comparative Example 1,” ginsenoside Rg3 is prepared

by “a typical method of separating an extract from ginseng with an alcohol extraction solvent” in which “20 g of ginseng crude-saponin was prepared and heated at 120°C for 3 hours.” *Id.* This example produces “2.01 g of ginsenoside (20S)-Rg3 and 1.7 g of ginsenoside (20R)-Rg3.” *Id.*

4. *Baek (Ex. 1008)*

Baek is titled “Dammarane Compound-Containing Pharmaceutical Composition for Renal Protection.” Ex. 1008, code (54). Baek’s pharmaceutical composition is used for a variety of ailments such as renal disease and diabetic nephropathy. *Id.* at Claims 1, 3. The composition includes a dammarane compound as an effective ingredient which can be ginsenoside Rk3 or ginsenoside Rh4. *Id.* ¶ 4. Baek explains that dammarane is a triterpenoid compound and is mainly found in *Panax* species in nature, such as *Panax ginseng*, but “is also found in *Gynostemma pentaphyllum* Makino, which is not a *Panax* species.” *Id.* ¶ 6. Baek discloses that “[w]hen heated, dammarane generates Rg2, Rg3, Rh2, and Rh1 through release of glycosyl residues and then generates Rg5, Rk1, Rh3, F4, and Rh4 through dehydration at position 20.” *Id.* ¶ 9. Baek further explains that it is known the main ingredients of the ginseng extract include ginsenosides Rg3, Rg5, and Rk1, which exhibit renal protective activities. *Id.* ¶ 13. Baek purports to provide a pharmaceutical composition that includes dammarane compounds not present in nature and can be used for renal protection. *Id.* ¶ 15.

Baek discloses that from a *Panax* species or a *Gynostemma pentaphyllum* Makino,

dammarane is heated to 50°C to 200°C, specifically 120°C, for 30 minutes to 10 hours, specifically 3 hours, followed by one or more, specifically three, filtration stages using reflux extraction in a solvent selected from the group consisting of lower alcohols or mixtures thereof, specifically CH₃OH, and then the resulting

filtrates are combined and concentrated under reduced pressure, thereby obtaining a CH₃OH extract, which, in turn, is dried under reduced pressure, suspended in water, and separated into a CH₂Cl₂ fraction, an n-BuOH fraction, and an H₂O fraction by sequentially using CH₂Cl₂ and water-saturated n-BuOH. Then, the fractions are dried by evaporation under reduced pressure, thereby obtaining CH₂Cl₂, n-BuOH, and H₂O extracts, respectively.

Id. ¶ 33. In another example, ginsenosides Rk3 and Rh4 are isolated by subjecting 1 kg of white ginseng to a heat treatment of 120°C for 3 hours, followed by three filtration stages using reflux extraction in CH₃OH for 3 hours. *Id.* ¶ 56.

5. *WO 178 (Ex. 1006)*

WO 178 “relates to a pharmaceutical composition and functional health food comprising an extract of processed *Panax* genus plant as an active ingredient for preventing and treating obesity and the use thereof.” Ex. 1006 ¶ 1. WO 178 explains that the plant belonging to the *Panax* genus has dammarane saponin as its most important ingredient and as an example, “Korean ginseng comprises high amount of ginsenosides, such as ginsenoside RbI, Rb2, Rc, Rd, RgI, Re and etc.” *Id.* ¶ 18. It is known to prepare a processed ginseng by high temperature and high pressure such that the main ginseng saponins, such as ginsenosides RbI, Rb2, Rc, and Rd, are changed into new saponin metabolites such as ginsenosides Rg3, Rg5, and RkI, that provide “new and more potent pharmacological effects.” *Id.* ¶ 19. One such *Panax* genus that has a root, stem, or leaf for this preparation is *Panax ginseng*. *Id.*

WO 178 purports to provide “a pharmaceutical composition comprising an extract of processed ginseng so as to make a ratio of ginsenoside (Rg3+Rg5+Rk1) to (Rb1+Rb2+Rc+Rd) of over 1.0, as an active

ingredient in an effective amount to treat and prevent obesity.” *Id.* ¶ 26. The composition may comprise a filtrate with a residue that is concentrated “to obtain [an] extract, as [the] active ingredient and pharmaceutically acceptable carrier, diluents or adjuvants to treat and prevent obesity.” *Id.* ¶ 32.

To make its composition, WO 178 discloses a process that includes autoclaving a dried plant material from the *Panax* genus at a temperature ranging from 70 to 150°C, preferably 100 to 130°C, for a period ranging from 2 to 6 hours, preferably 3 to 5 hours; mixing the material with 1 to 20-fold, preferably, 3 to 10-fold weight (kg) of water, C₁–C₄ lower alcohol, such as methanol, ethanol, butanol or the mixtures thereof, preferably ethanol; and then heating for the period ranging from 3 to 10 hours, preferably 3 to 6 hours, by reflux extraction with water, cold water extraction, ultra-sonication or conventional extraction. *Id.* ¶ 44.

E. *Obviousness in view of Kim 460 and Razmovski*

Petitioner contends that claims 1–13 are rendered obvious under 35 U.S.C. § 103(a) over Kim 460 and Razmovski. *See* Pet. 22–41; Reply 1–24. Patent Owner disagrees. PO Resp. 33–50; Sur-Reply 1–32. We address the parties’ contentions below.

1. *Petitioner’s Contentions*

Petitioner contends that “Kim 460 discloses that a ‘conventional use’ of its [ginseng] extractions is a medicinal agent.” Pet. 35 (citing Ex. 1005, 4:40–41). Kim 460 suggests “that ginseng is commonly used for treating hyperlipidemia and diabetes, which are metabolic syndromes.” *Id.* (citing Ex. 1005, 1:25–30). Specifically, Petitioner contends that Kim 460 “states that an extract [of ginseng] with increased levels of Rg3 and Rg5 is useful

for treating diabetes.” *Id.* (citing Ex. 1005, 9:8–25). “Kim 460 teaches extracting saponins from ginseng using ethanol as a solvent . . . Kim 460 heats ginseng or an extract of ginseng at 120°–180° C. for 0.5 to 20 hours.” Pet. 23 (citing Ex. 1005, 3:13–23, 3:48–49, 4:21–22).

Petitioner contends that Kim 460 teaches extracting saponins from ginseng using ethanol. Pet. 23 (citing Ex. 1005, 3:48–49). The text from Kim 460 relied on by Petitioner is reproduced below:

The solvent which may be employed herein includes a water, lower alcohol such as methanol, ethanol, etc., lower ketone such as acetone, methylethylketone, etc., supercritical fluid or mixed solvent thereof.

Ex. 1005, 3:48–51 (emphasis added).

Petitioner relies on Razmovski for teaching that *Gynostemma pentaphyllum* (GP) is a known equivalent for ginseng containing many of the same ginsenoside components. *See* Pet. 24 (citing Ex. 1003 ¶¶ 35 (“interchanging GP as a source of ginsenosides of Rb1, Rb2, Rc, and Rd which are chemically and pharmacologically identical to the ginsenosides in Kim 460”), 102). Razmovski “shows that ginsenosides Rb1, Rb2, Rc and Rd are present in [(*G. pentaphyllum*)], and observes that ginsenosides make up 25% of the saponin content of GP.” Pet. 24 (citing Ex. 1007, 4 (Table 1)). According to Petitioner, “Razmovski teaches that GP contains a significant and useful amount of the same ginsenosides that are used by Kim 460.” *Id.* “Razmovski teaches that saponins in GP have ‘biological activities congruent to the very popular expensive herbal medicine, *P. ginseng*.’” *Id.* (citing Ex. 1007, 18). “Razmovski [similarly] discusses numerous studies finding that GP is effective for modulating lipid metabolism and hyperglycaemia.” Pet. 35 (citing Ex. 1007, 4, 10–13). Razmovski teaches that “[t]he major interest of *G. pentaphyllum* originated from its lipid

lowering effects observed in animal models and human studies.” Ex. 1007, 18.

Based on the teachings in Kim 460 and Razmovski, Petitioner concludes that a person of ordinary skill in the art would have been

motivated to substitute GP for the ginseng (as taught by Razmovski) in the process of Kim 460: heating a GP extract to 120-180 °C in an autoclave for 0.5 to 20 hours to transform the Rb1, Rb2, Rc and Rd in the GP extract (as taught by Razmovski) to Rg3 and Rg5 and Rk1.

Pet. 24 (citing Ex. 1003 ¶ 102). According to Petitioner, “[h]eating in an autoclave elevates pressure, and heating to the temperatures of 120° to 180°C would increase the pressure to more than 1.2 atmospheres.” Pet. 23 (citing Ex. 1003 ¶ 41 (“The fact that heating a sealed vessel increases pressure is captured in the ideal gas law ($PV=nRT$).”). Petitioner concludes that “performing the process of Kim 460, using an extract of GP (as taught by Razmovski), must result in the same extract. Performing the same heating process, on the same product, produces the same results.” Pet. 26.

Petitioner further contends that applying heat and pressure are known techniques for increasing the concentrations of therapeutic agents in a biological extract. Reply 5 (citing Ex. 1005, 1:23–30, 6:34, 9:23–25; Ex. 1006 ¶¶ 68, 69, 93, 140). Here, the same source material Rb1, Rb2, Rc and Rd is shown by Razmovski to be present in GP. *Id.* (citing Ex. 1003 ¶¶ 35, 102); *see also* Reply 5 (“Patent Owner can hardly now dispute motivation. Patent Owner’s own expert suggested substituting GP for ginseng as a source of dammarane saponins in 2005, several years before the purported invention date. (Ex. 1022, 5).”). Petitioner concludes that it would have been obvious to one of ordinary skill in the art at the time the invention was made “to substitute GP for ginseng, as taught by Razmovski, in the

process of Kim 460, and that substitution would result in an extract with at least 0.7% of damulin A and 0.5% of damulin B.” Pet. 36.

2. *Patent Owner’s Contentions*

According to Patent Owner, “Petitioner admits that the prior art did not disclose the Claimed Concentrations and attempts to fill the missing gaps using inherent obviousness theory based on Experimental Reports together with a post-filing publication.” PO Resp. 13, *see also id.* at 33 (“Petitioner admits that neither Kim 460 nor Razmovski discloses or suggests DA, DB, or the Claimed Concentrations. Pet. 3, 18, 23, 25; Ex. 2015 ¶¶ 76–79; Ex. 2005, 211:21–212:3.”). “Petitioner’s challenge turns on proving that DA, DB, and their Claimed Concentrations are inherently present when Kim 460 is combined with Razmovski.” PO Resp. 34 (citing Pet. 3, 25). Patent Owner contends that “the Claimed Concentration limitations narrow the claims, and are not merely a ‘newly discovered function or property’ of the other limitations.” *Id.* In other words, Patent Owner contends that it is only by applying the claimed process method that the requisite concentrations of damulin A and damulin B are achieved.

Patent Owner further contends that Petitioner’s experimental reports (Ex. 1013 and Ex. 1016) fail to establish that the combination of Kim 460 with Razmovski achieves the claimed concentrations of damulin A and damulin B. PO Resp. 35. Specifically, Patent Owner contends that (a) “Petitioner did not adhere to methods disclosed in the prior art to obtain the experimental results” (*id.* at 35); (b) “[o]nly *some* of Petitioner’s results fall within the Claimed Concentrations” (*id.*); (c) “Dr. Lee admitted, the DB standard was impure, unreliable, and unstable” (*id.* at 36); (d) Petitioner cannot rely on Table 2 of the ’786 patent to establish inherency because

Petitioner's references fail to disclose or suggest the methods used to obtain those results disclosed in Table 2 of the '786 patent (*id.*); and (e) the post-filing reference relied on in the Petition also fails to support Petitioner's conclusion (*id.*).

3. Analysis

Kim 460 teaches that “heat-treatment of a ginseng may be performed at a temperature of 120° to 180°C. for 0.5 to 20 hours, preferably at a temperature of 120° to 140°C. for 2 to 5 hours . . . in an airtight container . . . [with] a small amount of water . . . preferably soaked in water and then heated in a closed container.” Ex. 1005, 3:21–35. Kim 460 teaches that water, methanol, and ethanol are suitable solvents. *Id.* at 3:48–51. Most of the methods exemplified in Kim 460 treat the ginseng with either steam or by adding water to the ginseng and then heating the ginseng in an airtight container. *See* Ex. 1005,¹² 5:1–13 (examples 1 and 2), 5:35–41 (Experimental Example 1).

Kim 460 teaches heat treating ginseng “at a temperature of 120° to 180°C. for 0.5 to 20 hours, preferably at a temperature of 120° to 140°C. for 2 to 5 hours. The heating time varies depending on the heating temperature.” Ex. 1005, 3:21–24. The extraction protocols in Kim 460 include: steaming ginseng in an autoclave (*id.* at 5:1–12 (Examples 1 and 2)), further

¹² Kim 460 exemplifies an alcohol extraction of ginseng using methanol, followed by autoclaving the sample at 120°C for 4 hours. *See* Ex. 1005, 5:15–22 (Example 3). Petitioner, however, does not appear to rely on this disclosure in Kim 460 in formulating their patentability arguments. *See generally* Pet. Patent Owner acknowledges that “Experiment 4 in Exhibit 1013 used a single autoclave heat-extraction step and used methanol as solvent like the examples in Kim 460, [but] the DA/DB results were well below the Claimed Concentrations.” PO Resp. 42.

processing the steamed ginseng with methanol under reflux (*id.* at 5:26–33 (Example 4)); adding water to ginseng followed by autoclaving (*id.* at 5:35–46 (Experimental Example 1)). Petitioner’s Declarant, Dr. Davies, explains that heating in a closed container, including an autoclave, follows the ideal gas law and would raise the pressure inside the container. *See* Ex. 1003 ¶ 41.

Kim 460 teaches that heat treating ginseng between 120 to 180°C significantly increases the Rg₃ and Rg₅ content compared to fresh ginseng. Ex. 1005, 6:65–7:3; *see also id.* at 6:49–64 (Table 2, showing highest Rg₃ and Rg₅ content at 130°C treatment for 2 hours), *see also id.* at 6:5–25 (Table 1 showing increase in Rg₃ and Rg₅ levels with extended heating). Kim 460 teaches that heat treated ginseng shows an increase in the vasodilation effect of the ginseng. *See* Ex. 1005, 9:9–22. Specifically, Kim 460 teaches that “activity of the processed ginseng having a ratio of ginsenoside (Rg₃+Rg₅) to (Rc+Rd+Rb₁+Rb₂) of above 1.0 according to the present invention was stronger about 50 fold than that of the original white ginseng extract which was not heat-treated at 50% vasodilation concentration.” Ex. 1005, 9:13–18.

Razmovski teaches that “the saponin content of the [*G. pentaphyllum*] plant has become the centre of extensive investigation which includes a variety of biological activities congruent to the very popular expensive herbal medicine, *P. ginseng*. Both contain dammarane-type glycosides, some of which are common to both plant species.” Ex. 1007, 18. Razmovski teaches that *G. pentaphyllum* is a more cost-effective source material for ginsenosides. *Id.*; *see also* Ex. 1022, 2 (“*G. pentaphyllum* could provide a variety of saponin types, an inexpensive source of saponins for pharmaceutical use and possibly, a substitute for *P. ginseng*”). In Table 1, Razmovski teaches that *G. pentaphyllum* contains the ginsenoside Rb₁, Rb₂,

Rd, Rc, and Rg₃ among the dammaranes listed. Ex. 1007, 9. Not only is *G. pentaphyllum* an art recognized equivalent for ginseng, Razmovski also suggests that the lower cost associated with *G. pentaphyllum* would further motivate the substitution. *Id.* at 18; *compare* Ex. 1005, 6:5–25 (Table 1 listing ginsenoside Rb₁, Rb₂, Rc, Rd, Rg₃, and Rg₅) *with* Ex. 1007, 9 (Table 1 listing ginsenoside Rb₁, Rb₂, Rc, Rd, and Rg₃). Thus, Razmovski teaches that ginsenosides Rb₁, Rc, Rb₃, Rg₃, malonyl-Rb₁ and malonyl-Rd are found in *G. pentaphyllum* and are the same as those in *P. ginseng*. *Id.* at 4.

Table 1 of Kim 460 is reproduced, again (for convenience), below:

TABLE 1							
Relative Peak Area of Saponin Components of Ginseng							
Sample	Saponin						(Rg ₃ + Rg ₅) / (Rc + Rd + Rb ₁ + Rb ₂)
	Rb ₁	Rb ₂	Rc	Rd	Rg ₃	Rg ₅	
130° C./2 hrs	4.06	3.44	3.98	3.82	21.00	16.17	2.43
120° C./3 hrs	10.82	6.91	8.52	7.64	28.01	14.01	1.24
120° C./2 hrs	10.46	7.74	10.66	5.58	24.12	11.35	1.03
110° C./2 hrs	19.02	11.55	10.68	8.37	7.02	4.39	0.23
White Ginseng	22.15	3.25	5.86	2.47	0.00	0.00	0.00
Red Ginseng	30.11	9.69	12.45	1.76	1.05	1.05	0.01

Table 1 of Kim 460 shows that upon extended heating under pressure the ginseng saponin components levels of Rb₁, Rb₂, Rc, and Rd decreased over time while the saponin component Rg₃ and Rg₅ increased. Ex. 1005, 6:5–25; *see also* Ex. 2015 ¶ 237.

Table 1 of Ex. 1019 is reproduced below:

Table 1. Content (w/w, %) of Ginsenosides in Raw Ginseng and Ginsengs Steamed at 100, 110, and 120 °C^a

	raw ginseng	steamed ginseng		
		100 °C	110 °C	120 °C
F ₄	n.d. ^b	0.14 ± 0.03	0.18 ± 0.02	0.23 ± 0.05
Rg ₂	0.13 ± 0.01	0.20 ± 0.02	0.32 ± 0.05	0.30 ± 0.03
Rg ₁	0.39 ± 0.02	0.35 ± 0.02	0.27 ± 0.04	0.22 ± 0.04
Rg ₅	n.d. ^b	0.15 ± 0.03	0.35 ± 0.05	0.64 ± 0.08
Rg ₃	n.d. ^b	0.24 ± 0.03	0.62 ± 0.06	1.32 ± 0.14
Rf	0.11 ± 0.02	0.12 ± 0.03	0.10 ± 0.04	0.10 ± 0.04
Re	0.38 ± 0.03	0.30 ± 0.04	0.08 ± 0.02	0.02 ± 0.01
Rd	0.28 ± 0.04	0.27 ± 0.03	0.20 ± 0.04	0.14 ± 0.08
Rc	0.65 ± 0.03	0.57 ± 0.05	0.40 ± 0.08	0.17 ± 0.06
Rb ₂	0.52 ± 0.03	0.44 ± 0.04	0.26 ± 0.04	0.10 ± 0.03
Rb ₁	0.56 ± 0.03	0.50 ± 0.03	0.30 ± 0.03	0.12 ± 0.03

^a Mean ± sd, *n* = 3). ^b n.d. = not detected.

Table 1 of Ex. 1019, shows a change in ginsenoside content upon heating ginseng. Ex. 1019, 2. The ginsenoside component levels of Rg₁, Rg₂, Rf, Re, Rd, Rc, Rb₁, and Rb₂ decreased with increasing temperature treatment. Ex. 1019, 2 (Table 1); Ex. 2015 ¶ 238 (showing a graph of the content of Table 1 from Ex. 1019, however, omitting the results from Rg₃ and Rg₅). We note that Table 1 of Ex. 1019, just like Kim 460, shows that upon heating ginseng, there is a concomitant increase in ginsenoside components Rg₃ and Rg₅ levels.

“Obviousness does not require absolute predictability of success . . . all that is required is a reasonable expectation of success.” *In re Droge*, 695 F.3d 1334, 1338 (Fed. Cir. 2012) (quoting *In re Kubin*, 561 F.3d 1351, 1360 (Fed. Cir. 2009) (citing *In re O’Farrell*, 853 F.2d 894, 903–04 (Fed.Cir.1988)); *Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359 (Fed. Cir. 2016) (explaining that the expectation of success issue involves a showing of “a reasonable expectation of achieving *what is claimed*”) (emphasis added).

Here, Petitioner relies on Kim 460 to show that heating ginseng, which contains the ginsenosides Rb₁, Rb₂, Rc, Rd, Rg₃, and Rg₅, results in a

product that has increased levels of the Rg₃ and Rg₅ components. Kim 460 teaches that ginseng with the higher Rg₃ and Rg₅ content is 50-fold more potent in their vasodilation studies than the non-treated samples. Razmovski teaches that *G. pentaphyllum* contains over 100 saponins and eight of those saponins are the same as protopanaxadiol-type ginsenosides Rb₁, Rc, Rb₃, Rd, F₂, Rg₃, malonyl-Rb₁, and malonyl-Rd that are found in *P. ginseng*. Ex. 1007, 4.

Knowing that *G. pentaphyllum* also contains ginsenosides Rb₁, Rb₂, Rc, Rd, Rg₃, and Rg₅, we find Petitioner's conclusion that heating a *G. pentaphyllum* extract that contains these ginsenosides would similarly increase the ginsenoside levels of Rg₃ and Rg₅ is a reasonable inference from the cited references. Based on the teachings in Kim 460, we agree with Petitioner that it would have been reasonable to expect that the same source material, although derived from a different plant, would behave similarly to heat treatment. See Pet. 24. Because *G. pentaphyllum* is a less expensive starting material, we also agree with Petitioner that there is sufficient motivation to substitute *G. pentaphyllum* for ginseng in the methods of Kim 460.

Neither Kim 460 nor Razmovski suggests the presence of damulin at any level. "[I]nherency may supply a missing claim limitation in an obviousness analysis." *PAR Pharm., Inc. v. TWI Pharm., Inc.*, 773 F.3d 1186, 1194 (Fed. Cir. 2014). "[T]he concept of inherency must be limited when applied to obviousness, and is present only when the limitation at issue is the 'natural result' of the combination of prior art elements." *Id.* at 1195.

Inherency may supply a missing claim limitation in an obviousness analysis. An inherent characteristic of a formulation can be part of the prior art in an obviousness analysis even if the inherent characteristic was unrecognized or unappreciated by a

skilled artisan. It is long settled that in the context of obviousness, the mere recitation of a newly discovered function or property, inherently possessed by things in the prior art, does not distinguish a claim drawn to those things from the prior art. The Supreme Court explained long ago that it is not invention to perceive that the product which others had discovered had qualities they failed to detect.

Persion Pharms. LLC v. Alvogen Malta Operations Ltd., 945 F.3d 1184, 1189 (Fed. Cir. 2019) (citations, internal quotations marks, brackets and parentheticals omitted).

In an effort to support an inherency argument, Petitioner provides experimental data and expert declarations. *See* Pet. 25–27; Ex. 1003; Ex. 1013 (Experimental Report-1);¹³ Ex. 1016 (Experimental Report-2); Ex. 1035. We next address the sufficiency of these disclosures.

a) Experimental Methods

Patent Owner contends that “Petitioner’s evidence is deficient and fatally flawed because Petitioner and its declarants admitted that the experiments did not adhere to the methods disclosed in the prior art, used an impure and unstable DB standard, and, despite using hindsight knowledge of the goal and conditions necessary to achieve it, produced results falling both inside and outside the Claimed Concentrations.” PO Resp. 13. Patent Owner argues that Petitioner’s results are, therefore, not the necessary natural result of the combination of elements explicitly disclosed in the prior art. *Id.* at 39 (citing Ex. 2015 ¶¶ 174–184).

Patent Owner contends that Petitioner’s experimental protocols did not follow the methods in Kim 460 and Razmovski. PO Resp. 45. Because

¹³ We refer to Exhibit 1013 and the “Experimental Report-1” interchangeably throughout this Decision.

Petitioner's experimental methods deviated significantly from the prior arts disclosure, Patent Owner contends that they are, therefore, not probative of inherency. PO Resp. 45 (citing *Glaxo Group Ltd. v. Apotex, Inc.*, 376 F.3d 1339, 1349 (Fed. Cir. 2004)).

Patent Owner contends that

Case-dispositive admissions that Ex. 1013 did not follow the prior art, and instead added results-affecting steps and conditions not suggested by the references, and that the results fell inside and outside the Claimed Concentrations, and were calculated with a low purity reference standard and were unreliable, and manipulated by omitting calibration points and unsubstantiated calibration line corrections, mean that the Ex. 1013 results have little to no probative value and cannot be used to reach a legal conclusion of inherency.

Sur-Reply 3–4.

After considering the parties' arguments and evidence developed during the full trial, we are persuaded by Patent Owner's arguments and determine that Petitioner has not shown by a preponderance of the evidence that based on the combination of Kim 460 and Razmovski that claims 1–13 of the '786 patent are unpatentable. We find that Petitioner has not provided sufficient evidence from which to conclude that the application of Kim 460's methods as applied to the *G. pentaphyllum* starting material of Razmovski would necessarily result in the damulin levels as claimed.

The protocol disclosed in Experimental Report-1 submitted by Petitioner applied water to the plant material and then raised the temperature to 130°C for 3 hours in an autoclave achieving a pressure of 1.56. mPa. See Ex. 1013, 10, 12–13. The extraction protocols of *G. pentaphyllum* actually used in the Experimental Report-1 (described in the section "4. Results") differs from that described in section 3. of the report titled "Experimental

Design”. *Compare* Ex. 1013, 10 (3. Experimental Design) *with id.* at 12–13 (4. Result). The experiments applied to obtain the “Results” of the Experimental Report-1 utilize a second autoclave step and not the reflux extraction as described in the Experimental Design of the report. *Compare* Ex. 1013, 10 (3. Experimental Design) *with id.* at 12–13 (4. Result). The samples in Experimental Report-1 were then assayed for the following compounds: Gypenoside L, Gypenoside LI, Ginsenoside Rg3, Damulin A, and Damulin B. Ex. 1013, 26 (Tables 1–4).

Kim 460 suggests heat treatment temperature in a range of 120–180°C for 0.5 to 20 hours. Ex. 1005, 3:21–23. Kim 460’s protocols fall within the following categories:

- (1) autoclave and steaming at 120°C for 4 hours (Ex. 1005, 5:2–6 (Example 1); PO Resp. 41 (“In Examples 1 and 2, Kim 460 performed a single autoclave step and did not perform any extraction. Ex. 2015 ¶ 176; Ex. 1005, 5:1–13.”);
- (2) autoclave and steaming at 130°C for 2 hours (Ex. 1005, 5:9–12 (Example 2));
- (3) autoclave and steaming at 120°C for 4 hours followed by methanol reflux (Ex. 1005, 5:26–33 (Example 4); PO Resp. 41–42 (“Example 4 processed ginseng obtained from Example 1 by reflux extraction (at unknown temperatures) using methanol in a water bath) for 4 hours. Ex. 2015 ¶ 176; Ex. 1005, 5:28–32.”);
- (4) adding sample and water to an airtight stainless-steel vessel and heating 110°C for 2 hours, 120°C for 2 hours or 3 hours, and 130°C for 3 hours, followed by a methanol extraction (Ex. 1005, 5:35–6:25 (Table 1));

- (5) heat treating at 100°C, 110°C, 120°C, 130°C, 150°C, 160°C, 180°C, and 200°C for 2 hours (Ex. 1005, 6:25–65 (Table 2)), *see also id.* at 7:10–37 (utilizing a 4-hour heat treatment);
- (6) extraction in methanol under reflux in a water bath for 4 hours followed by placing the sample into an autoclave and heating at 120°C for 4 hours before drying the sample at 60°C (Ex. 1005, 5:15–24 (Example 3)).

We agree with Patent Owner that Kim 460 does not apply a second autoclave step in any of the disclosed extraction protocols in contrast to the experiments presented by Petitioner. *See* PO Resp. 45; *see generally* Ex. 1005; Ex. 1013, 12–13; *see* Pet. 26–27.

The protocols in Experimental Report-1 (Exhibit 1013) relied on by Petitioner to establish the requisite damulin A and damulin B levels based on the combination of Kim 460 and Razmovski, therefore, do not follow the protocols set out in Kim 460. Sur-Reply 1 (Petitioner failed to faithfully follow the prior art “admitting that . . . experiments neither followed any combination of the prior art nor showed the Claimed Concentrations were necessarily present therein.”), *see also id.* at 1–2 (Petitioner asserting that “performing a second autoclaving step ‘would not affect the results’ (Reply 13). This contradicts Davies’ statement that the second autoclaving exposed samples to 5- to 8-fold higher pressure than heating without autoclaving, and heating for an additional 4 hours in solvent. Ex. 1035 ¶ 42; Ex. 2017:168:18–169:20”).

Claim 1 is a method claim, not a product-by-process claim, and requires subjecting an *ethanol extract* of *G. pentaphyllum* leaves to a temperature of about 40–125°C. Ex. 1001, 16:63-17:2. Patent Owner contends that Experimental Report-1 (Exhibit 1013) “cannot be used to

establish inherency of [the method of] claim 1 of the '786 patent . . . [because t]he methods described in Ex. 1013 do not correspond to the limitations of the method of claim 1." PO Resp. 45–46 (citing Ex. 2015 ¶¶ 185–192). Specifically, claim 1 requires an *ethanol extract* of *G. pentaphyllum* that is then subjected to the heat treatment under pressure. PO Resp. 12–13. Because none of the experiments in Experimental Report-1 apply an ethanol extraction step before applying the heat treatment under pressure, they are not probative for establishing inherency of claim 1. *See* Ex. 1001, 16:60–17:2 (claim 1). We agree with Patent Owner that Experiments 1–3 in the Experimental Report-1 do not utilize an alcohol extraction step.

Experiment 4 in the Experimental Report-1 comes closer to the method of claim 1 because it uses an alcohol extract – methanol extract – that is then heated 60°C. *See* Ex. 1013, 26 (Table 4 - GP-M-01 and GP-M-02), *see id.* at 13 (describing that the extract of GP-M-01 and GP-M-02 are obtained by “[p]ut 50g of GP into a 2L beaker and add 600 g of Methanol and seal tightly. Place the bottle into the autoclave and heat at 60°C for 4 hours”). There is no evidence in the report, however, that this treatment step is under pressure. The damulin levels in that experiment are 0.17–0.18 % for damulin A and 0.18-0.2% for damulin B which are below the levels recited in claim 1 of the '786 patent. *See* Ex. 1013, 26 (Table 4); *see* Ex. 1001, 16:60–17:2 (claim 1).

Patent Owner urges us to disregard Dr. Davies’ conclusion that the temperature difference between the upper temperature limit of the '786 patent claims – 125°C – and the temperature of 130°C used in Experimental Report-1 (Ex. 1013) is insignificant. PO Resp. 46 (citing Ex. 2015 ¶¶ 185–192; Ex. 1003 ¶ 45; 37 C.F.R. § 42.65(a) (“Expert testimony that

does not disclose the underlying facts or data on which the opinion is based is entitled to little or no weight.”). Here, Kim 460 suggests a temperature range 120–180°C that overlaps with the temperature range of about 40–125 C as recited in claim 1 of the ’786 patent, yet the Experimental Report-1 set the temperature at 130°C which is above the temperature recited in claim 1.

Patent Owner contends that “Kim 460 demonstrates that increasing the temperature by 10 degrees (e.g. 120°C→130°C) increased Rg3 and Rg5 concentrations by 5-fold.” PO Resp. 47 (citing Ex. 2015 ¶ 190; Ex. 1005, 6:48–65 (Table 2)). We agree with Patent Owner that there is sufficient evidence in the record to support the position that increasing the temperature would increase the Rg3 and Rg5 levels. PO Resp. 47 (“Wu demonstrates that increasing the temperature by 10 degrees (e.g., 120°C (3h)→130°C (3h)) increases DA/DB concentrations by 4 to 5-fold. Ex. 1012, Table 2; Ex. 2015 ¶191.”). This indicates that temperature is a result effective variable for manipulating the Rg3 and Rg5 product levels.

We agree with Patent Owner that the disclosure in Experimental Report-1 (Exhibit 1013) is insufficient to establish the damulin concentrations of claim 1 of the ’786 patent as necessarily being the “natural result” of applying the methods of Kim 460. Here, the Experimental Report-1 (Ex. 1013) does not utilize an ethanol extract of *G. pentaphyllum*, does not follow any of the experimental protocols set out in Kim 460, and does not apply a temperature within the range recited in claim 1 of the ’786 patent. Based on these deficiencies, we agree with Patent Owner that the disclosure in the Experimental Report-1 (Ex. 1013) is not sufficient to establish that the damulin levels as recited in claim 1 of the ’786 patent are the natural result of Kim 460’s methods.

Even though claims 8 and 10 do not have the same temperature, pressure, and solvent constraints as claim 1, we agree with Patent Owner's expert "that the experimental methods used in Exhibit 1013 used a second autoclaving step and that performing a second autoclaving step was not disclosed in Petitioner's prior art references." Ex. 2015 ¶ 145 (citing Ex. 2006, 105:16–18; 109:1–7; Ex. 2005, 114:2–115:1; 161:20–162:4). Because the methods in the Experimental Report-1 deviate from the disclosure of the references relied upon in the Petition, we agree with Patent Owner that the Experimental Report-1 (Exhibit 1013) is insufficient to establish that the damulin levels are necessarily the "natural result" of applying the methods of Kim 460 to *G. pentaphyllum* starting material.

b) Results Unreliable / Impure Standard

Patent Owner contends that Petitioner's use of a highly impure damulin B standard makes the results unreliable. *See* PO Resp. 16. "The amount of DB in the experimental GP samples in Exhibit 1013 cannot be accurately or reliably quantified because the DB standard used to prepare the calibration curve was too impure, produced a double peak on all of the chromatograms, and was unstable." *Id.* (citing Ex. 2006, 22:10–24; 60:5–15; 64:8–65:16; 67:8–15). In Experimental Report-2 (Ex. 1016) the damulin B levels could not be determined because "the standard used for damulin B was too low to determine the content, and a chromatography of damulin B sample showed a double peak." PO Resp. 16 (citing Ex. 1016, 2; Ex. 1017 ¶ 19); Ex. 2015 ¶ 145. Patent Owner's expert, "Dr. Naumovski explains that because there are one or more impurities that co-elute with the DB standard, it is not possible to accurately measure the peak areas that correlate only to

DB because one will inevitably also include some portion of the overlapping impurities.” *Id.* at 17 (citing 2015 ¶¶ 143, 241).

The damulin B standard used in the Experimental Report-1 did not come with “a certificate of analysis to verify the purity and identity of the DB standard, [where] as it did for DA, the manufacturer provided the ‘Sample Information’ showing that the DB standard has a purity of 74% and contained other compounds.” PO Resp. 17–18 (citing Ex. 2015 ¶¶ 146–154, 243; Ex. 1013, 6–7). According to Patent Owner, “Dr. Lee also admitted that the DB standard used in Exhibits 1016 and 1013 was unstable and the purity decreased over time.” PO Resp. 17 (citing Ex. 2006, 22:10–24:1; 24:17–24), *see id.* at 18 (citing Ex. 2015 ¶¶ 142–143, 234, Ex. 2006, 64:8–65:16; 22:20–23:11; 60:5–15). Furthermore, “Dr. Lee testified that he used 70% as a correction factor in all of the calibration data for DB, but again did not disclose this fact or provide any data or experiments showing the 70% purity.” PO Resp. 18 (citing Ex. 2015 ¶ 241; Ex. 2006, 65:1–16); Ex. 2010, 5–7, 15–17.

Petitioner contends that “[w]hen using a standard with known amounts of impurity, it is appropriate to correct the calibration curve to account for the impurity. Reply 7 (citing Ex. 1035 ¶¶ 20 (“It is not necessary to use a 100% pure standard to create a reliable calibration curve.”), 32). Petitioner also contends that shoulders and split peaks are very common in HPLC procedures. Reply 12 (citing Ex. 1035 ¶ 53).

Patent Owner’s expert, Dr. Naumovski explains that overlapping peak areas makes calculation unreliable. PO Resp. 19 (citing 2015 ¶ 161); Ex. 2015 ¶¶ 142 (“a double peak is shown for damulin B in each of the damulin B chromatograms on each of pages 16-22 of Exhibit 1013 and also throughout Exhibits 1016, 1033, and 1034”); 159 (citing Ex. 2007).

“If overlapping peaks are symmetric (Gaussian) and of equal size, there is no area error by integration. If they are of non-equal size, the smaller one is too small, the large too large, irrespective of their order. If overlapping peaks are tailed (the usual case), the area of the first one is too small, the second one is too large.” Ex. 2007, 319. Figure 2 of Exhibit 2007 is reproduced below:

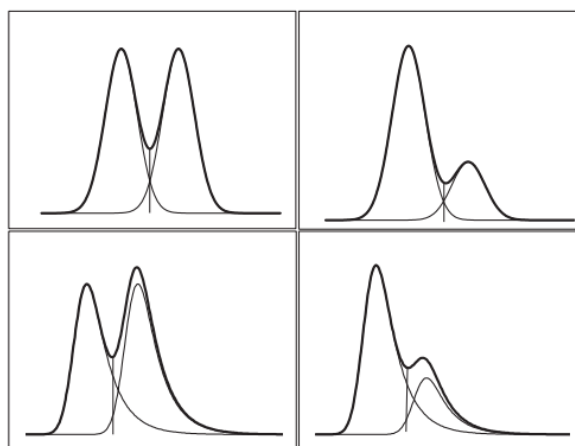
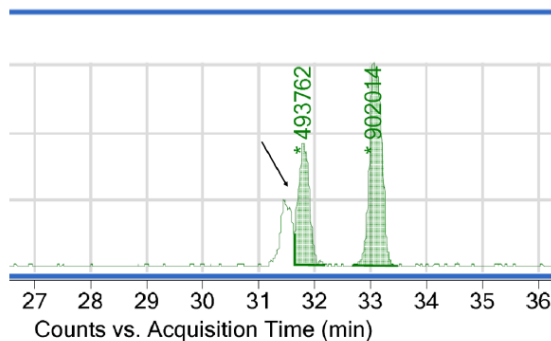


Figure 2 shows analyte distribution in overlapping chromatographic peaks. Ex. 2007, 320 (Figure 2). “Integration errors when overlapped chromatographic peaks are separated by a vertical drop. Top left: Both peaks are integrated correctly because they are of equal size and symmetrical. Top right: The area of the small peak is too small, irrespective of the elution order. Bottom [both panels]: In the case of tailing, the area of the second peak is too large.” *Id.* “The case of ‘rider peaks’ [see top and bottom right Figure 2 above], i.e. of a small peak sitting on the tailing edge of a much larger one, is a topic of itself.” *Id.* at 319.

Experimental Report-1 provides chromatograms of the analyzed samples. *See* Ex. 1013, 14–22. A blow up of GP-WW-01 (3rd trial) in Exhibit 1013 is reproduced below:



The figure shows an asymmetric double peak. *See* Ex. 2015 ¶ 157 (citing Ex. 1033, 5; Ex. 1013, 15).

Patent Owner’s expert, “Dr. Naumovski explains that, without completely resolving the peaks, the amount of DB determined to be in the experimental extracts is not accurate or reliable.” PO Resp. 19 (citing 2015 ¶ 161); Ex. 2015 ¶ 159 (citing Ex. 2007). “These impurities and overlapping peak areas make the calculations of the peak areas, and resultant calibration curves and concentrations in the samples unreliable as Dr. Lee testified.” PO Resp. 20 (citing Ex. 2015 ¶ 143; Ex. 1016, 2; Ex. 1017 ¶ 19; Ex. 2006, 63:20–25; 64:8–12, 88:1–7, 96:2–97:2, 97:8–23).

Petitioner contends that the objections raised by Dr. Naumovski with respect to the experimental results as shown in Experimental Report-1 (Ex. 1013), have been reviewed by Dr. Davies and he concluded that each of those objections do not raise a legitimate cause for concern. Reply 21 (citing (Ex. 1035 ¶¶ 26–32, 50–52). “Dr. Davies has testified that the practices used by Dr. Lee for the extraction and heat treatment processes used equipment and processes that were common and conventional at the filing date of the ‘786 patent and would be obvious in view of the references.” Reply 23 (citing Ex. 1035 ¶ 52).

According to Patent Owner, however, Dr. Davies’ testimony establishes he was not informed sufficiently about the experimental

problems, and the evidence required for establishing inherent obviousness. Sur-reply 10–11 (citing Ex. 2005, 110:3–7; Ex. 2017, 16:12–19, 22:18–22, 51:7–59:8, 62:6–63:14, 64:2–66:3, 115:6–11, 115:20–24, 119:4–6, 175:12–17). Patent Owner contends that Petitioner’s Reply and Dr. Davies’ “second declaration fail to provide persuasive reasoning or evidence that the Claimed Concentrations were necessarily present in the prior art . . . but admit that only some of the experiments . . . fell within the Claimed Concentrations, while some results indisputably fell outside the Claimed Concentrations.” Sur-Reply 5.

After considering the parties’ arguments and evidence developed during the full trial, we determine that Petitioner has not shown by a preponderance of the evidence that applying the methods of Kim 460 to the *G. pentaphyllum* starting material of Razmovski necessarily results in the damulin levels as claimed. We agree with Dr. Naumovski’s explanation that the use of an impure damulin B standard containing a double peak renders the results as shown in Exhibit 1013 unreliable. Ex. 2015 ¶¶ 141–143, 162–173; *see* Ex. 2007.

Dr. Naumovski further explains that the use of an impure standard would require the application of a correction factor, but this correction factor would need to be reassessed as the purity of the standard decreases over time. *See* Ex. 2015 ¶ 166. As Dr. Naumovski explains that applying a correction factor to the linear regression would result in a shallower slope. Ex. 2015 ¶ 165 (citing 2006, 71:12–73:5; 99:2–9). “Reducing the calibration curve slope increases the amount of damulin B calculated from the experimental extracts.” Ex. 2015 ¶ 165. However, in the Experimental Report-1 the same correction factor for the damulin B standard was used to

quantify the measurements taken on different days. Ex. 2015 (citing Ex. 2006, 22:10–23:11; 23:24–24:1; 60:5–20; 64:13–65:16; 67:8–15).

Finally, we also agree with Patent Owner’s expert that there was no reason to rely on impure damulin B standard when high purity damulin B standards were commercially available and should have been used to conduct experiments especially when trying to *quantify* the amounts of a particular compound, in this case damulin B. *See* Ex. 2015 ¶¶ 150–151.

c) Damulin Levels Insufficient

Patent Owner’s expert testifies that “a POSA cannot reliably quantify the amount of damulin B in the experimental extracts in Ex. 1013.” Ex. 2015 ¶ 157; PO Resp. 20 (Ex. 2015 ¶¶ 157–158; Ex. 2006, 63:23–25; 97:8–23). Here, “Dr. Lee [Petitioner’s expert] admitted that to quantify the peak areas, and thereafter calculate the amount of DB extracted, Petitioner drew a line straight down between the overlapping portion of the double peak in each instance.” PO Resp. 20–21 (citing Ex. 2015 ¶ 159; Ex. 1013, 15–22; Ex. 1033, 4–15; Ex. 2006, 96:2–10). Patent Owner contends that asymmetric peaks cannot be accurately or reliably quantified this way. PO Resp. 21 (citing Ex. 2015 ¶¶ 159–160; Ex. 2007, 320).

For the reasons discussed above (*see* II.E.3.b), we agree with Patent Owner that when the calculation is based on a tailing asymmetric peak, the area of the second peak is too large, thus reliance on such calculations is improper. Ex. 2007, 320 (Figure 2); Ex. 2015 ¶ 157 (citing Ex. 1033, 5; Ex. 1013, 15).

Here, “Petitioner does not dispute that Experiments 1–3[□] in Ex. 1013 used methods not explicitly disclosed by prior art” and therefore the results are not “the natural result of the combination of elements explicitly

disclosed by the prior art, which is fatal to Petitioner’s challenges.” Sur-Reply 6 (internal footnote omitted). Patent Owner contends, and we agree, that “[f]or experiments to prove the Claimed Concentrations were inherent, they must adhere to methods explicitly disclosed by prior art.” Sur-Reply 8 (citing *Par Pharm., Inc.*, 773 F.3d at 1195-96; *Eli Lilly & Co. v. Barr Labs., Inc.*, 251 F.3d 955, 970 (Fed. Cir. 2001) (inherency must “flow[] from the reference’s explicitly explicated limitations”)).

According to Patent Owner, Petitioner used different methods to quantify the damulin B standard and the purported damulin B sample in the experimental *G. pentaphyllum* extracts. According to Patent Owner’s expert, Petitioner used the entire double peak to quantify “for the DB standard at each concentration to prepare the calibration curves for Trial 1 and Trial 2” and then applied the correction factor (0.7) to the whole peak area. PO Resp. 21 (citing 2015 ¶¶ 162–163; Ex. 1033, 1, 3; Ex. 2015 ¶ 163; Ex. 2010, 1, 5). Petitioner’s expert “Dr. Lee acknowledged that applying the 0.7 correction factor would change the linear regression equation and result in a shallower slope.” PO Resp. 22 (citing Ex. 2015 ¶ 165; Ex. 2006, 71:12–73:5; 99:2–9; Ex. 2010). “[R]educing the calibration curve slope increases the DB amount calculated from the experimental extracts.” PO Resp. 22 (citing Ex. 2015 ¶¶ 165–166; Ex. 1013, 24). In essence, manipulating the correction factor is like putting one’s thumb on a scale to get a desired outcome.

Patent Owner’s expert further noted that Dr. Lee conducted only two experiments several days apart without using a reliable internal standard to validate the quantification. PO Resp. 25 (citing Ex. 2015 ¶¶ 171–172; Ex. 2012, 225). “[T]he GP-**-02 results are systematically lower than GP-**-01, but it cannot be determined which of GP-**-01 and GP-**-02 is the outlier, and there is no intra-day measurement, or measurements within

shorter number of days to validate the results.” PO Resp. 24 (citing Ex. 2015 ¶ 169). A person of ordinary skill in the art would “commonly perform[] experiments at least in triplicates to have a basic understanding of the statistical variation” and to figure out which of the two results is an outlier PO Resp. 26 (citing Ex. 2015 ¶ 195).

After considering the parties’ arguments and evidence developed during the full trial, we determine that Petitioner has not shown by a preponderance of the evidence that applying the methods of Kim 460 to the *G. pentaphyllum* starting material of Razmovski results in the requisite damulin levels as claimed. We agree with Dr. Naumovski’s explanation that the use of an impure damulin B standard containing a double peak render the results as shown in Exhibit 1013 unreliable. Ex. 2015 ¶¶ 141–143.

Dr. Naumovski further explains that the use of impure standard would require the application of a correction factor, but this correction factor would need to be reassessed as the purity of the standard decreases. *See id.* ¶ 166. However, the same correction factor for the damulin B standard was used for quantifying the measurements taken in the Experimental Report-1 (Ex. 1013) on different days. Ex. 2015 (citing Ex. 2006, 22:10–23:11; 23:24–24:1; 60:5–20; 64:13–65:16; 67:8–15). “[W]hen correcting calibration curves, the calculated DB concentration can shift either inside or outside the Claimed Concentrations.” Sur-Reply 12 (citing Ex. 2015 ¶ 165); Ex. 2015 ¶ 159 (quoting Ex. 2007, 319) (“to avoid peak area integration errors ‘all peaks to be quantified should be fully separated in a chromatogram.’”). Based on the full record, we agree with Patent Owner’s position that “the low purity DB standard led to inaccurate and unreliable quantification of DB concentrations.” Sur-Reply 12 (citing Ex. 2015 ¶¶ 140–206).

d) Reliance on the '786 Specification

Patent Owner contends that “Petitioner cannot show inherency by relying on results in Table 2 of the '786 patent because Petitioner’s references fail to disclose or suggest [the same] methods used to obtain those results [set out in Table 2].” PO Resp. 36 (citing Ex. 2015 ¶¶ 331–332).

Petitioner contends that “the methodology for creating TG1022 is not part of the claims nor does Patent Owner suggest that it contributes to patentability. In any event, creating an extract from plant material using a lower alcohol such as ethanol as a solvent, heating, and then evaporating the supernatants is taught by” prior art references. Reply 3. Petitioner contends that “[t]he heating conditions in Table 2 are consistent with Kim 460 and WO 178. Table 2 discloses heating at 121 °C for 0.5 to 24 hours, and the heating is performed in an autoclave.” *Id.* Because these heating parameters fall within the conditions taught by Kim 460 and WO 178, Petitioner contends that the information contained in Table 2 of the '786 patent can, therefore, be relied on to show the inherent properties of the prior art methodologies. *See* Reply 2–4.

The data in Table 2 of the '786 patent was obtained from a specific method described therein, which is not taught in any combination of Petitioner’s references. PO Resp. 47–48 (citing Ex. 1001: 3:49–4:3; Ex. 2015 ¶ 331).

The extraction process as set out in the '786 patent is as follows:

[D]ried *Gynostemma pentaphyllum* leaves are extracted with an ethanol aqueous solution having a volume 10 to 20 times as large as the volume of the leaves (preferably, a 20~80% ethanol aqueous solution), and a first supernatant is taken out. To the residual ethanol-dipped material of the *Gynostemma pentaphyllum* leaves, an ethanol aqueous solution (6 to 20 times) (preferably, a 20~80% ethanol aqueous solution) is again added,

and then a second supernatant is taken out through 1-8 hour extraction at 65~95° C. Then, the first and second supernatants are put together and are filtrated by using gauze, and the solution obtained through the filtration is centrifuged (800~1,500xG) to remove floating matter. The resulting extracted product is concentrated to Brix concentration of 40 to 70 by vacuum-concentration to provide a *Gynostemma pentaphyllum* extract concentrate (hereinafter, referred to as ‘TG 1022’), and then the TG1022 is treated with high temperature and high pressure to provide a novel *Gynostemma pentaphyllum* extract concentrate having increased contents of damulins (hereinafter, referred to as ‘TG1022F’).

Ex. 1001, 3:52–4:3. Patent Owner pointed out that the data expressed in Table 2 of the ’786 patent started with an *ethanol extract* that is then subjected to high pressure and high temperature. PO Resp. 48. Kim 460 relied on by Petitioner extracts ginseng root using either steam or water under heat and pressure, and further teaches that the extract may additionally be processed with alcohol under reflux. *See* Ex. 1005, 3:20–34; 5:1–25, 5:35–7:10. The process in Kim 460, therefore, differs from the process recited in Table 2 as well as claim 1 of the ’786 patent which start with an *ethanol extract* that is then heat treated under pressure. *See* Ex. 1001, 16:63–65.

Here, “Petitioner never alleges or establishes the Claimed Concentrations are necessarily present in an extract of GP heated under conditions explicitly disclosed by Kim 460.” PO Resp. 38 (citing Pet. 27; Ex. 2015 ¶¶ 174–184).

All of Petitioner’s experiments deviated in multiple ways from what is disclosed in Kim 460 and Naumovski—they used two autoclaving steps, used different ratios of solvents to GP in each step, did not subject an ethanol extract of GP leaves to heating

(40-125°C) and high pressure, and did not obtain the Claimed Concentrations in some of the experiments.

PO Resp. 37.

Patent Owner contends, and we agree, that Petitioner has not established that the claimed concentration of damulin are the natural result of applying the conditions disclosed in Kim 460. PO Resp. 38 (citing Ex. 2015 ¶¶ 174–184, 193–200; Ex. 1003 ¶ 85; Ex. 2005, 147:12–20; Ex. 1013, 26; Ex. 1014, Table 4); Ex. 1003 ¶ 85 (“The only result that was outside of the range of damulin A and damulin B in claim 10 is one of the two experiments that simply used water as a solvent.”); Ex. 2005, 147:12–20 (Dr. Davies testifies “there are some results that are below the claimed range, and there are some results that are within the claim range.”).

We agree with Patent Owner that “Kim 460 only uses a single autoclave heat-treatment step and does not provide any suggestion to conduct two autoclaving heat-extraction steps.” PO Resp. 40 (citing Ex. 2015 ¶ 176). “[B]y deviating from the prior art and performing two consecutive autoclaving steps, Petitioner’s protocols [in the Experimental Report -1 (Ex. 1013)] were intended to extract more DA/DB than by performing one autoclave extraction as disclosed in the prior art.” PO Resp. 40 (citing Ex. 2006, 33:19–25). We also agree with Patent Owner, that “Petitioner fails to establish a nexus between any particular prior art disclosures and the specific procedures and materials used to obtain [the ’786 patent] Table 2’s data.” Sur-Reply 20–21.

e) Post Filing Date Reference

Patent Owner contends that the post-filing reference—Wu—relied on in the Petition also fails to support Petitioner’s conclusion that applying the

methods of Kim 460 to *G. pentaphyllum* will necessarily and naturally result in damulin levels as claimed. PO Resp. 36.

Extrinsic evidence can be used to demonstrate what is “necessarily present” in a prior art embodiment even if the extrinsic evidence is not itself prior art. *See Monsanto Tech. LLC v. E.I. DuPont de Nemours & Co.*, 878 F.3d 1336, 1345 (Fed. Cir. 2018) (allowing “non-prior art data” to be used to support inherency); *Schering Corp. v. Geneva Pharm., Inc.*, 339 F.3d 1373, 1377 (Fed. Cir. 2003) (finding that the prior art need not recognize the inherent property).

“Wu does not disclose how the samples are heat processed and, therefore, [Wu] cannot be relied on to establish inherency.” PO Resp. 47 (citing Dec. 29). Wu discloses that “[t]he leaves of crude and heat-processed *G. pentaphyllum* (5.0 g) were extracted with 50 mL of 80% methanol by ultrasonication for 1.5 h and then evaporated.” Ex. 1012, 311. Wu shows that extracts from “*G. pentaphyllum* heat processed at 130°C for 3 h [results in damulin A content of 11902 µg/g, damulin B content of 5877 µg/g], whereas those from the raw plant were below 250 µg/g. Longer treatment times (>3 h) did not give any significant changes in the saponin contents.” *Id.* at 315. Because it cannot be established based on Wu’s disclosure that the samples are autoclaved, just like the samples in Kim 460, Wu cannot be relied on to establish inherency of Kim 460’s process when applied to *G. pentaphyllum* as the starting material.

f) Conclusion

Having considered the arguments and evidence based on the full record, we determine that Petitioner has not sufficiently established by a preponderance of the evidence that the damulin A and B levels are the

natural result of the combination Kim 460 and Razmovski. Therefore, Petitioner has not shown by a preponderance of the evidence that claims 1–13 of the '786 patent are unpatentable as obvious over Kim 460 and Razmovski.

F. *Obviousness in view of Kim 460 and Kang or Baek*

Petitioner contends that claims 1–13 are rendered obvious under 35 U.S.C. § 103 over Kim 460 and Baek or Kang. *See* Pet. 41–47. Patent Owner disagrees. PO Resp. 50–56.

Petitioner contends that Kim 460 describes overlapping process conditions for making extracts of ginseng. Pet. 41–42 (citing Ex. 1005, 3:13–35; Ex. 1003 ¶ 41).

According to Petitioner, “Kang is directed to a pharmaceutical composition with Rg5 and Rk1 as active ingredients. (Ex. 1010, claim 1). Kang creates Rg5 and Rk1 by heating an alcohol extract at 130°C for 2 hours. (*Id.* at 7). Kang explicitly teaches using GP as an alternative to ginseng to create Rg5 and Rk1. (*Id.* at 4).” Pet. 42. Petitioner contends that “the GP heating described by Kang and Baek is typically performed in an autoclave, it would be obvious to use the same instrument to control heating of a solvent extraction.” Reply 14 (citing Ex. 1035 ¶¶ 48–49). Additionally, Petitioner contends that “the pressures of the second step are too low to produce any measurable difference.” Reply 15 (citing Ex. 1035 ¶¶ 40–44).

Kang states:

Ginsenoside Rg5 and Rk1, . . . may be purified and separated through column chromatography by processing the roots of *Panax* plants containing ginseng saponin, for example, *Panax ginseng* . . . or *Gynostemma pentaphyllum* containing a compound of the dammaran[e] backbone, and the like, leaves of

these plants, tissue cultures of these plants, or extracts thereof with water or lower alcohols.

Ex. 1010, 4.

Petitioner relies on Baek for teaching “heating GP at a temperature of 50° to 200°C for 0.5 to 20 hours and extracting saponins from the heated GP using alcohol as a solvent.” Pet. 42 (citing Ex. 1008 ¶ 33). “Baek also discloses that GP is a substitute for ginseng as a source of dammarane saponins.” Pet. 43 (citing Ex. 1008 ¶¶ 6, 33). Petitioner concludes that it would have been obvious to one of ordinary skill in the art to substitute GP for ginseng in Kim 460 based on the teachings of Kang and Baek. Pet. 43.

Petitioner relies on the Experimental Report-1 to establish that applying the heat treatment method of Kim 460 results in the recited damulin content. Experimental Report-1 shows experiments performed under the direction of Petitioner’s Declarant, Dr. Davies, to establish “that heating [GP] at 130°C for 3 hours produces an extract with damulin levels that exceed the amounts of claim 8 [and fall with the range of claim 10] using several different solvents.” Pet. 43 (citing Ex. 1013 (Tables 2–3); Ex. 1012, 315 (Table 2)).

Patent Owner presents similar arguments to those discussed with respect to the combination of Kim 460 and Razmovski. *See* PO Resp. 33–50. For the reasons discussed above (*see II.E*), we agree with Patent Owner that “Petitioner and its declarants failed to perform any experiments that meet the requirements of the Challenged Claims and failed to establish that combining Kim 460 and Baek or Kang rendered the Challenged Claims inherently obvious.” PO Resp. 51. “Kang actually discloses extracting ginseng using methanol, then extracting further using ether three times, then further extracting using a mixture of butanol and ethyl acetate three times,

then extracting with butanol three times, and then heating the butanol extract at 130°C for 2 hours.” PO Resp. 51 (citing Ex. 2015 ¶ 265; Ex. 1010, 7).

Because “Kang does not disclose pressure at all and instead subjects ginseng to a multi-step extraction process using a series of solvents and solvent mixtures, none of which is ethanol, and at a higher temperature than the claimed range,” we agree with Patent Owner that Kang does not remedy the deficiencies of Kim 460. PO Resp. 51–52 (citing Ex. 2015 ¶ 265). We also agree with Patent Owner, that “[n]one of the references provides any disclosure, data, or suggestion for making an ethanol extract of GP leaves and subjecting an ethanol extract of GP (or ginseng) leaves to high pressure and high temperature.” PO Resp. 55.

Having considered the arguments and evidence based on the full record, we determine that Petitioner has not sufficiently established by a preponderance of the evidence that the damulin A and B levels are the necessary and natural result of the combination Kim 460 and Kang or Baek. Therefore, Petitioner has not shown by preponderance of the evidence that claims 1–13 of the ’786 patent are unpatentable as obvious over Kim 460 and Kang or Baek.

G. *Obviousness in view of WO 178 and Razmovski*

Petitioner contends that claims 8–10, 12, and 13 are rendered obvious under 35 U.S.C. § 103 over WO 178 and Razmovski. *See* Pet. 47–55. Patent Owner opposes. PO Resp. 56–58.

Petitioner contends that WO 178 “discloses heating ginseng at 70 to 150°C in an autoclave for 2 to 6 hours to convert dammarane saponins Rb1, Rb2, Rc and Rd into Rg3, Rg5 and Rk1, and extracting the converted saponins from the ginseng using an alcohol solvent.” Pet. 48 (citing

Ex. 1006 ¶¶ 19, 26, 44). WO 178 performs heating in an autoclave, which would result in increased pressure. Pet. 49 (citing Ex. 1006 ¶ 44; Ex. 1003 ¶ 41 (“The fact that heating a sealed vessel increases pressure is captured in the ideal gas law ($PV=nRT$).”).

Petitioner relies on Razmovski for teaching that *G. pentaphyllum* includes several of the same saponins as ginseng, therefore, rendering obvious the substitution of *G. pentaphyllum* for ginseng. See Pet. 48–49 (citing Ex. 1004 ¶ 36). Petitioner relies on the teachings of Razmovski to establish “that GP includes several of the same saponins as ginseng including Rb1, Rb2, Rc and Rd, and make up 25% of the saponin content of GP.” Pet. 48 (citing Ex. 1007, 4 (Table 1)). Petitioner contends that “an extract of GP that has been heated within the parameters described by WO 178, would have at least one of 0.7% of damulin A or 0.5% of damulin B.” Pet. 49.

Based on these combined teachings, Petitioner concludes that a person of ordinary skill in the art “would have a reasonable expectation of success for this substitution [of *G. pentaphyllum* for ginseng in WO 178’s process] since Razmovski shows that the source materials in WO 178, Rb1, Rb2, Rc and Rd, are all present in GP.” Pet. 49 (citing Ex. 1003 ¶ 36).

Patent Owner argues that Petitioner has not provided a sufficient basis from which to conclude that the claimed composition and extract based on the combination of WO 178 and Razmovski are inherently obvious. See PO Resp. 56 (citing Ex. 2016 ¶ 287).

WO 178 teaches producing an extraction of ginseng.

[D]ried plant material of Panax genus was cut into small pieces and the piece was autoclaved at the temperature ranging from 70 to 150°C, preferably 100 to 130°C, for the period ranging from 2 to 6 hours, preferably 3 to 5 hours; and was mixed with 1 to

20-fold, preferably, 3 to 10-fold weight (kg) of water, C₁-C₄ lower alcohol, such as methanol, ethanol, butanol or the mixtures thereof, *preferably ethanol*; was heated for the period ranging from 3 to 10 hours, preferably 3 to 6 hours, by reflux extraction with water, cold water extraction, ultra-sonication or conventional extraction, preferably by reflux extraction with water; the residue was filtered and then the filtrate was dried at the temperature ranging from 40 to 80°C, preferably from 50 to 70°C, to obtain inventive extract of processed ginseng.

Ex. 1006 ¶ 44 (emphases added); *see also id.* ¶ 118 (“1 kg of dried root of *Panax ginseng* was cut into small pieces and the sliced pieces were heated at 120°C for 4 hours in autoclave. The processed ginseng was mixed with 2 *I* [sic] of ethanol and heated for 4 hours by reflux extraction with water.”). Ginseng processed at 120°C for 4 hours in an autoclave shows increased Rg3, Rg5, Rg1 content and a reduction of Rb1, Rb2, Rc, and Rd. *Id.* ¶ 127 (Table 1).

Petitioner’s Declarant, Dr. Davies, selected a treatment protocol for Experimental Report-1 that falls within the preferred treatment temperature and time disclosed in WO 178. *See* Ex. 1003 ¶¶ 104 (Explaining that “the experimental conditions of temperature, pressure and extraction solvent are within the processing conditions described by Kim 460, WO 178, and Baek.”), 105–107; Ex. 1006 ¶ 44; Ex. 1013. Experimental Report-1 shows that damulin A and B are within the claimed concentrations. *See* Ex. 1013, 26; Ex. 1003 ¶ 105 (“As expected, the use of methanol or ethanol with reflux extraction provided higher contents of all compounds analyzed compared to use of a simple water extraction alone The results are what I expected based on the prior art.”). According to Dr. Davies, results showing higher compound content in the examples that use methanol and ethanol extraction is expected. Ex. 1003 ¶ 105.

Patent Owner presents similar arguments to those discussed with respect to the combination of Kim 460 and Razmovski. *See* PO Resp. 56–58. For the reasons discussed above (*see supra* Section II.E), we agree with Patent Owner that Experimental Report-1 (Ex. 1013), “in addition to not following the steps of WO 178 (alone or in view of Razmovski), also failed to establish inherent results of the claimed concentrations of damulin A and damulin B.” PO Resp. 58. Patent Owner further contends that the results of Experimental Report-1 are flawed because “Dr. Lee admitted that the same impure, unstable, and unreliable reference standard and double peaks were used and present.” *Id.*

To make its composition, WO 178 discloses a process that includes autoclaving pieces of dried *Panax ginseng* at a temperature of 120°C for 4 hours; the processed ginseng is then mixed with ethanol and heated for 4 hours by reflux extraction with water. Ex. 1006 ¶¶ 116–118. While the extraction protocols of *G. pentaphyllum* in the Experimental Report-1 utilize two autoclave steps and do not use a reflux extraction. Ex. 1013, 12–13 (4. Result). The Experimental Report-1, therefore, does not follow the protocol set out in WO 178.

We agree with Patent Owner, that by not following the steps of WO 178 alone or in view of Razmovski, and using an impure damulin B standard, the results obtained in the Experimental Report-1 (Ex. 1013) are not sufficient to establish that the damulin B level is the “natural result” of applying the method of WO 178 to the *G. pentaphyllum* starting material Razmovski. For the reasons discussed above (*see supra* Section II.E), we agree with Dr. Naumovski’s explanation that the use of an impure damulin B standard containing a double peak render the results as shown in Exhibit 1013 and 1016 unreliable. Ex. 2015 ¶¶ 141–143. Dr. Naumovski further

explains that the use of impure standard would require the application of a correction factor, but this correction factor would need to be reassessed as the purity of the standard decreases. *See id.* ¶ 166. However, the same correction factor for the damulin B standard was used for quantifying the measurements taken in the Experimental Report-1 (Ex. 1013) on different days. *See* Ex. 2015 ¶ 167 (citing Ex. 2006, 22:10–23:11; 23:24–24:1; 60:5–20; 64:13–65:16; 67:8–15). “[W]hen correcting calibration curves, the calculated DB concentration can shift either inside or outside the Claimed Concentrations.” Sur-Reply 12 (citing Ex. 2015 ¶¶ 165–166); Ex. 2015 ¶ 159 (quoting Ex. 2007, 319) (“to avoid peak area integration errors ‘all peaks to be quantified should be fully separated in a chromatogram.’”).

Dr. Naumovski explains that applying a correction factor of 0.7 to the linear regression would result in a shallower slope. Ex. 2015 ¶ 165 (citing 2006, 71:12–73:5, 99:2–9). “Reducing the calibration curve slope increases the amount of damulin B calculated from the experimental extracts.” *Id.* We, therefore, agree with Patent Owner that the record supports the position that “the low purity DB standard led to inaccurate and unreliable quantification of DB concentrations.” Sur-Reply 12 (citing Ex. 2015 ¶¶ 140–206).

Having considered the arguments and evidence based on the full record, we determine that Petitioner has not sufficiently established by a preponderance of the evidence that the damulin A and B levels are the natural result of the combination WO 178 and Razmovski. Therefore, Petitioner has not shown by preponderance of the evidence that the claims 8–10, 12, and 13 of the ’786 patent are unpatentable as obvious over WO 178 and Razmovski.

H. *Obviousness in view of WO 178 and Kang*

Petitioner contends that claims 1–13 are rendered obvious under 35 U.S.C. § 103 over WO 178 and Kang. *See* Pet. 55–63. Patent Owner disagreed. PO Resp. 58–60.

Patent Owner contends that “Kang fails to remedy the deficiencies of WO178, . . . as it not only fails to disclose or suggest the claimed method, but it actually discloses a different extraction procedure for ginseng and the combination therefore leads the POSA in a different direction than WO178 and the ’786 patent’s method.” *Id.* at 59 (citing Ex. 2015 ¶¶ 298–299); *see* Dec. 40. Specifically, “Kang fails to provide motivation to subject an ethanol extract of GP leaves to high pressure and high temperature.” *Id.* (citing Ex. 2015 ¶ 311).

We agree with Patent Owner that Petitioner has not sufficiently explained why based on the teachings of Kang one would have substituted methanol for ethanol in Kang’s extraction method. Based on that apparent deficiency, with respect to Kang’s extraction method, Petitioner has not made a sufficient showing that the Petition demonstrates that there is a reasonable likelihood that claims 1–13 are unpatentable over WO 178 and Kang.

III. MOTION TO STRIKE

In its Motion, Patent Owner asserts that the supplemental second declaration of Dr. Neal Davies (Ex. 1038) was untimely filed on June 5, 2023. Paper 35. Patent Owner asserts that our rules only permit a party to “serve” supplemental evidence, but not to file it. *Id.* (citing 37 C.F.R. § 42.64(b)(2)). Patent Owner asserts that Petitioner does not oppose the motion to strike. *Id.* (citing Ex. 3003).

According to our rules, “[s]upplemental evidence is not filed at the time of the objection, but simply served, and is filed only in support of an opposition to a motion to exclude.” CTPG¹⁴ 79 (citing 37 C.F.R. §§ 42.64(a), 42.64(b)(2)).

We recognize that striking the entirety or a portion of evidence is generally an exceptional remedy. CTPG, 80. This is because the Board is generally capable of identifying new issues or belatedly-presented evidence when weighing the evidence at the close of trial and disregarding any new issues or belatedly-presented evidence that exceeds the proper scope of Reply or Sur-Reply. *Id.*

However, in this case, Petitioner has not explained why the filing of the supplemental second declaration of Dr. Neal Davies was necessary. Petitioner also does not explain why any information contained in this supplemental second declaration could not have been included in the second Davies declaration (Ex. 1035) filed with Petitioner’s Reply. Petitioner also did not file an opposition to a motion to exclude, thus, Exhibit 1038 was not filed in support of an opposition to a motion to exclude. Finally, Patent Owner asserts that Petitioner does not oppose the Motion to Strike. Paper 35 (citing Ex. 3003).

Under the facts and circumstances of this proceeding we find striking the supplemental second declaration of Dr. Neal Davies (Exhibit 1038) is justified.

¹⁴ PTAB *Consolidated Trial Practice Guide*, November 2019 (the “CTPG”), available at: <https://www.uspto.gov/about-us/news-updates/consolidated-trial-practice-guide-november-2019>.

IV. CONCLUSION

For the foregoing reasons, we conclude that Petitioner has not established by a preponderance of the evidence that claims 1–13 of the ’786 patent are unpatentable.

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–13	103	Kim 460, Razmovski		1–13
8–10, 12, 13	103	WO 178, Razmovski		8–10, 12, 13
1–13	103	Kim 460, Kang, Baek		1–13
1–13	103	WO178, Kang		1–13
Overall Outcome				1–13

V. ORDER

Accordingly, it is

ORDERED that Petitioner has not shown by a preponderance of the evidence that claims 1–13 of the ’786 patent are unpatentable;

FURTHER ORDERED that Patent Owner’s Motion to Strike the supplemental second declaration of Dr. Neal Davies is *granted*;

FURTHER ORDERED that Exhibit 1038 shall be struck from the record;

FURTHER ORDERED that, because this is a final written decision, parties to this proceeding seeking judicial review of our decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2022-00998
Patent 8,357,786 B2

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